

BEC303 Network Analysis — syllabus extract

Extracted verbatim from 3ecsy11.pdf (238,451 characters of machine-readable text).

in the theory SEE component.

10

Network Analysis

Course Code

1BEC303

Scheme

2025

Type of Course

PCC

Semester

III

Teaching Hours/Week (L:T:P: S)

3:2:0

CIE Marks

50

Total Hours of Pedagogy (

L

:T:P:

SL

&

TW

)

42:28:0:50

SEE Marks

50

Credits

4

Total Marks

100

Examination type (SEE)

Theory

Exam Hours

3

Course outcome (Course Skill Set) At the end of the course, the student will be able to:

1.

Analyze electronic networks using mesh and nodal analysis techniques for DC and AC circuits with dependent and independent sources.

2.

Evaluate electronic circuit responses (currents, voltages, and power) using network theorems.

3.

Analyze RL, RC, and RLC circuits under transient conditions using initial

and final condition concepts.

4.

Analyze linear electronic circuits using Laplace transform techniques.

5.

Evaluate the performance of two-port networks by determining parameters and impedance

characteristics.

Module-1

Basic Concepts- Mesh and Node Analysis:

Introduction, Source of Electrical Energy, Kirchhoff's Laws, Source transformations, General Network Transformations, Mesh and Node analysis with linearly dependent and independent sources for DC and AC networks. Text book 1: 1.1, 3.2, 3.3, 4.1-4.4, 5.2, 5.5,

10.1-

10.6

Number of Hours: 15

Module-2

Network Theorems for DC and AC circuits:

Introduction, Superposition Theorem, Reciprocity Theorem, Conditions for Reciprocity, Examples of Reciprocal/Non-Reciprocal circuits, Thevenin's Theorem, Maximum Power transfer theorem. Case Study

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1: Maximum Power Transfer in Electronic Systems Design.

Text book 1: 5.1, 5.3, 5.4, 10.7; Textbook 2: 4.9

Number of Hours: 13

Module

-3

Transient behavior and initial conditions:

Behavior of circuit elements under switching condition and their Representation, Evaluation of initial and final conditions in RL, RC and RLC circuits for AC and DC excitations.

Case Study: RC Circuit

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Based Power-ON Delay in Electronic Indicator Systems.

Text book 1: 7.1, 7.2, 7.7, 8.1

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8.9, 9.1-9.7, 10.1-10.7

Number of Hours: 14

Module

-4

Laplace Transformation & Applications:

Solution of networks, step, ramp and impulse responses, Waveform Synthesis, Transfer function of elementary passive circuits, Network Synthesis Case Study: Concept of poles for first order passive filters and their use in computing time response characteristics and bandwidth. Text book 1: 14.3

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14.7, 15.1-15.8; Textbook 2: 6.8.

Number of Hours: 14

Module-5

Two -Port Network:

Introduction, Characterisation of Linear Time-invariant Two-port Networks,

Open- Circuit

Impedance Parameters, Short- circuit Admittance Parameters, Transmission Parameters,

Short introduction to Hybrid parameters (1 hour material), Why multiple representations are required, Two-port

Symmetry, Input impedance in terms of Two-port Parameters, Output Impedance. Text book 1:

17.1

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17.6

Number of Hours: 14

11

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals): Text books:

1.

Hayt, Kemmerly and Durbin-Engineering Circuit Analysis, TMH 8th Edition, 2012

2.

Network Analysis

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A Simplified Approach, K Channa Venkatesh and D Ganesh Rao, Cengage India Private

Limited,

2012

,

Second Edition, ISBN-13: 978-8131533215

Reference books / Manuals:

1.

M.E.Van Valkenburg (2000), Network Analysis, Prentice Hall of India, 3rd edition, 2000,

ISBN:9780136110958.

2.

J.David Irwin/ R.Mark Nelms- Basic Engineering Circuit Analysis John Wiley, 8th ed, 2006.

3.

Charles K Alexander and Mathew NO Sadiku-Fundamentals of Electric Circuits, Tata Page 17 BE III Semester

in ECE/TCE 15.09.2023 McGraw-Hill, 3rd Ed, 2009.

Web links and Video Lectures (e-Resources):

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<https://nptel.ac.in/courses/108105159>

Teaching-Learning Process (Innovative Delivery Methods): The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1.

Ensure at least 2 hours of review of mathematical concepts such as matrix algebra, complex variables, first and second order differential equations during the tutorial sessions

2.

Present case studies for each module which have industry relevance

3.

Use of simulation tool or even hardware demonstrations in the classroom to illustrate complex concepts

Questions from the Case Study topics shall not appear in the theory SEE component